

Exponential Equations and an Introduction to Logarithms (Part II)

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{10 \cdot 3} = x^{30}$

Strange New Exponent Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$, $x \neq 0$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Equations in which Two Bases Must Be Changed

Solve. Write the Roman numeral of each exponent rule that you use.

① $8^{x+10} = 4^{3x}$

$(2^3)^{x+10} = (2^2)^{3x}$

II $2^{3(x+10)} = 2^{2(3x)}$ VIII

$2^{3x+30} = 2^{6x}$

$3x+30 = 6x$

$30 = 3x$

$10 = x$

② $25^{4-x} = \frac{1}{125^x}$

$25^{4-x} = 125^{-x}$ IX

$(5^2)^{4-x} = (5^3)^{-x}$

$5^{2(4-x)} = 5^{3(-x)}$ V

$5^{8-2x} = 5^{-3x}$

$8-2x = -3x$

$x = -8$

③ $\sqrt[4]{27} = 81^x$

$27^{\frac{1}{4}} = 81^x$

$(3^3)^{\frac{1}{4}} = (3^4)^x$

$3^{3(\frac{1}{4})} = 3^{4(x)}$

$3^{\frac{3}{4}} = 3^{4x}$

$\frac{3}{4} = 4x$

$\frac{3}{16} = x$

④ $9^{2x-5} = 27^x$

$(3^2)^{2x-5} = (3^3)^x$

II $3^{2(2x-5)} = 3^{3(x)}$ VIII

$3^{4x-10} = 3^{3x}$ V

$4x-10 = 3x$

$x = 10$

⑤ $\frac{1}{16} = 64^{x-1}$

$\frac{1}{4^2} = 64^{x-1}$ IX

$4^{-2} = 64^{x-1}$

$4^{-2} = (4^3)^{x-1}$ V

$4^{-2} = 4^{3(x-1)}$

$4^{-2} = 4^{3x-3}$

$-2 = 3x-3$

$\frac{1}{3} = x$

⑥ $32^{4x} = \sqrt{8}$

$32^{4x} = 8^{\frac{1}{2}}$

$(2^5)^{4x} = (2^3)^{\frac{1}{2}}$

$2^{5(4x)} = 2^{3(\frac{1}{2})}$

$2^{20x} = 2^{\frac{3}{2}}$

$20x = \frac{3}{2}$

$x = \frac{3}{40}$

More Complicated Exponential Equations

Solve. Write the Roman numeral of each exponent rule that you use. If an answer is irrational, round to the nearest ten thousandth.

$$\textcircled{7} 3(2^{5x-1}) + 4 = 28 \quad \textcircled{8} -10 = \frac{7^x}{-9} + 1 \quad \textcircled{9} -1 = 8(3^{2-4x}) - 57$$

$$3(2^{5x-1}) = 24$$

$$2^{5x-1} = 8$$

$$2^{5x-1} = 2^3$$

$$5x-1 = 3$$

$$\boxed{x = \frac{4}{5}}$$

$$\textcircled{8} -10 = \frac{7^x}{-9} + 1$$

$$-11 = \frac{7^x}{-9}$$

$$99 = 7^x$$

$$\log_7 99 = x$$

$$\boxed{2.3614 \approx x}$$

$$\textcircled{9} -1 = 8(3^{2-4x}) - 57$$

$$56 = 8(3^{2-4x})$$

$$7 = 3^{2-4x}$$

$$\log_3 7 = 2-4x$$

$$\log_3 7 - 2 = -4x$$

$$\frac{\log_3 7 - 2}{-4} = x$$

$$\boxed{0.0572 \approx x}$$

$$\textcircled{10} 54 = 4 + 10(5^{8-4x}) \quad \textcircled{11} 3 + 4(9^x) = 35 \quad \textcircled{12} 8 + \frac{7^{-6x-3}}{-4} = -2$$

$$50 = 10(5^{8-4x})$$

$$5 = 5^{8-4x}$$

$$5^1 = 5^{8-4x}$$

$$1 = 8-4x$$

$$\boxed{\frac{7}{4} = x}$$

$$4(9^x) = 32$$

$$9^x = 8$$

$$x = \log_9 8$$

$$\boxed{x \approx 0.9464}$$

$$\frac{7^{-6x-3}}{-4} = -10$$

$$7^{-6x-3} = 40$$

$$-6x-3 = \log_7 40$$

$$-6x = \log_7 40 + 3$$

$$x = \frac{\log_7 40 + 3}{-6}$$

$$\boxed{x \approx -0.816}$$

$$\textcircled{13} 8 - 2(4^{\frac{x}{3}+5}) = -1$$

$$-2(4^{\frac{x}{3}+5}) = -9$$

$$4^{\frac{x}{3}+5} = \frac{9}{2}$$

$$\frac{x}{3} + 5 = \log_4 \left(\frac{9}{2}\right)$$

$$\frac{x}{3} = \log_4 \left(\frac{9}{2}\right) - 5$$

$$(3) \frac{x}{3} = (\log_4 \left(\frac{9}{2}\right) - 5)(3)$$

$$x = 3 \log_4 \left(\frac{9}{2}\right) - 15$$

$$\boxed{x \approx -11.7451}$$

$$\textcircled{14} 5 = 6(8^{\frac{x}{5}+7}) - 14$$

$$19 = 6(8^{\frac{x}{5}+7})$$

$$\frac{19}{6} = 8^{\frac{x}{5}+7}$$

$$\log_8 \left(\frac{19}{6}\right) = \frac{x}{5} + 7$$

$$\log_8 \left(\frac{19}{6}\right) - 7 = \frac{x}{5}$$

$$5(\log_8 \left(\frac{19}{6}\right) - 7) = \frac{x}{5} (5)$$

$$5 \log_8 \left(\frac{19}{6}\right) - 35 = x$$

$$\boxed{-32.2284 \approx x}$$

Problems Involving Quadratic Equations

Review on Solving Quadratic Equations by Factoring

Example I) $x^2 + 11x + 18 = 0$

$$(x+9)(x+2) = 0$$

$$\boxed{x = -9} \text{ or } \boxed{x = -2}$$

Example II) $0 = x^2 - 7x - 8$

$$0 = (x-8)(x+1)$$

$$\boxed{x = 8} \text{ or } \boxed{x = -1}$$

Example III) $x^2 - 10x + 21 = 0$

$$(x-7)(x-3) = 0$$

$$\boxed{x = 7} \text{ or } \boxed{x = 3}$$

Example IV) $0 = 3x^2 + x - 2$

$$0 = (3x-2)(x+1)$$

$$\boxed{x = \frac{2}{3}} \text{ or } \boxed{x = -1}$$

Solve. Write the Roman numeral of each exponent rule that you use. If an answer is irrational, round to the nearest ten thousandth.

⑮ $3(5^{-x^2+2x}) + 2 = 17$

$$3(5^{-x^2+2x}) = 15$$

$$5^{-x^2+2x} = 5$$

$$5^{-x^2+2x} = 5^1$$

$$-x^2 + 2x = 1$$

$$0 = x^2 - 2x + 1$$

$$0 = (x-1)(x-1)$$

$$\boxed{x = 1} \text{ or } \boxed{x = 1}$$

⑯ $-2(7^{x^2+x}) + 40 = -58$

$$-2(7^{x^2+x}) = -98$$

$$7^{x^2+x} = 49$$

$$7^{x^2+x} = 7^2$$

$$x^2 + x = 2$$

$$x^2 + x - 2 = 0$$

$$(x+2)(x-1) = 0$$

$$\boxed{x = -2} \text{ or } \boxed{x = 1}$$

⑰ $64 = 4^{x^2} \cdot 4^{2x}$

$$64 = 4^{x^2+2x}$$

$$4^3 = 4^{x^2+2x}$$

$$3 = x^2 + 2x$$

$$0 = x^2 + 2x - 3$$

$$0 = (x+3)(x-1)$$

$$\boxed{x = -3} \text{ or } \boxed{x = 1}$$

⑱ $\frac{2^{x^2}}{2^{4x}} = \frac{1}{16}$

$$2^{x^2-4x} = \frac{1}{16}$$

$$2^{x^2-4x} = 2^{-4}$$

$$2^{x^2-4x} = 2^{-4}$$

$$x^2 - 4x = -4$$

$$x^2 - 4x + 4 = 0$$

$$(x-2)(x-2) = 0$$

$$\boxed{x = 2}$$

⑲ $3^{3x} \cdot 3^{x^2} = \frac{1}{9}$

$$3^{3x+x^2} = \frac{1}{9}$$

$$3^{3x+x^2} = \frac{1}{3^2}$$

$$3^{3x+x^2} = 3^{-2}$$

$$3x + x^2 = -2$$

$$x^2 + 3x + 2 = 0$$

$$(x+1)(x+2) = 0$$

$$\boxed{x = -1} \text{ or } \boxed{x = -2}$$

⑳ $7 = \frac{7^{2x}}{7^{x^2}}$

$$7 = 7^{2x-x^2}$$

$$7^1 = 7^{2x-x^2}$$

$$1 = 2x - x^2$$

$$x^2 - 2x + 1 = 0$$

$$(x-1)(x-1) = 0$$

$$\boxed{x = 1} \text{ or } \boxed{x = 1}$$

Problems Requiring Substitution

② $3^{2x} - 5(3^x) + 4 = 0$

V $(3^x)^2 - 5(3^x) + 4 = 0$

Let $3^x = y$

$(y)^2 - 5(y) + 4 = 0$

Quadratic Equation $y^2 - 5y + 4 = 0$

$(y-1)(y-4) = 0$

$y = 1$ or $y = 4$

$3^x = 1$ or $3^x = 4$

VII $3^x = 3^0$ or $x = \log_3 4$

$x = 0$ or $x \approx 1.2619$

② $0 = 3(6^{2x}) - 4(6^x) - 15$

V $0 = 3(6^x)^2 - 4(6^x) - 15$

Let $6^x = y$

$0 = 3(y)^2 - 4(y) - 15$

Quadratic Equation $0 = 3y^2 - 4y - 15$

$0 = (3y+5)(y-3)$

$y = -\frac{5}{3}$ or $y = 3$

$6^x \neq -\frac{5}{3}$ or $6^x = 3$

$x = \log_6 3$

$x \approx 0.6131$

② $0 = 9^{2x} - 14(9^x) + 45$

V $0 = (9^x)^2 - 14(9^x) + 45$

Let $9^x = y$

$0 = (y)^2 - 14(y) + 45$

Quadratic Equation $0 = y^2 - 14y + 45$

$0 = (y-9)(y-5)$

$y = 9$ or $y = 5$

$9^x = 9$ or $9^x = 5$

VI $9^x = 9^1$ or $x = \log_9 5$

$x = 1$

$x \approx 0.7325$

② $4(8^{2x}) + 4(8^x) - 35 = 0$

$4(8^x)^2 + 4(8^x) - 35 = 0$

Let $8^x = y$

$4(y)^2 + 4(y) - 35 = 0$

Quadratic Equation $4y^2 + 4y - 35 = 0$

$(2y+7)(2y-5) = 0$

$y = -\frac{7}{2}$ or $y = \frac{5}{2}$

$8^x \neq -\frac{7}{2}$ or $8^x = \frac{5}{2}$

$x = \log_8 \left(\frac{5}{2}\right)$

$x \approx 0.4406$

Equations Requiring the Quadratic Formula

Review on Solving Quadratic Equations with the Quadratic Formula

If $ax^2+bx+c=0$, then $x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

Example V) $5x^2-3x-7=0$

$a=5, b=-3, c=-7$

$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$x = \frac{-(-3) \pm \sqrt{(-3)^2-4(5)(-7)}}{2(5)}$

$x = \frac{3 \pm \sqrt{9+140}}{10}$

$x = \frac{3 \pm \sqrt{149}}{10}$

Example VI) $0=-6x^2+x+3$

$a=-6, b=1, c=3$

$x = \frac{-1 \pm \sqrt{1^2-4(-6)(3)}}{2(-6)}$

$x = \frac{-1 \pm \sqrt{73}}{-12}$

$x = \frac{1 \pm \sqrt{73}}{12}$

Solve. Write the Roman numeral of each exponent rule that you use. If a logarithm is irrational, round to the nearest ten thousandth.

25) $72 = 8 + 2(2^{x^2+5x})$

$64 = 2(2^{x^2+5x})$

$32 = 2^{x^2+5x}$

$2^5 = 2^{x^2+5x}$

$5 = x^2 + 5x$

Quadratic Equation. Standard Form

$0 = x^2 + 5x - 5$

$x = \frac{-b \pm \sqrt{b^2-4ac}}{2a}$

$x = \frac{-5 \pm \sqrt{5^2-4(1)(-5)}}{2(1)}$

$x = \frac{-5 \pm \sqrt{45}}{2}$

$x = \frac{-5 \pm 3\sqrt{5}}{2}$

26) $0 = 5(8^{2x}) - 4(8^x) - 3$

$0 = 5(8^x)^2 - 4(8^x) - 3$

Let $8^x = y$

Quadratic Equation. $0 = 5y^2 - 4y - 3$

$y = \frac{-(-4) \pm \sqrt{(-4)^2-4(5)(-3)}}{2(5)}$

$y = \frac{4 \pm \sqrt{76}}{10}$

$8^x = \frac{4 \pm \sqrt{76}}{10}$

$8^x = \frac{4 + \sqrt{76}}{10}$ or $8^x = \frac{4 - \sqrt{76}}{10}$

$x = \log_8\left(\frac{4 + \sqrt{76}}{10}\right)$

$x \approx 0.1156$

$$\textcircled{27} \frac{5^{7x^2}}{5^{2x}} = 125$$

$$\text{II} \quad 5^{7x^2-2x} = 125$$

$$5^{7x^2-2x} = 5^3$$

Quadratic
Equation
Standard
Form

$$7x^2 - 2x = 3$$

$$7x^2 - 2x - 3 = 0$$

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(7)(-3)}}{2(7)}$$

$$x = \frac{2 \pm \sqrt{88}}{14}$$

$$x = \frac{2 \pm 2\sqrt{22}}{14}$$

$$x = \frac{1 \pm \sqrt{22}}{7}$$

$$\text{IV} \quad \textcircled{28} -7^{2x} + 10(7^x) + 2 = 0$$

$$-(7^x)^2 + 10(7^x) + 2 = 0$$

$$\text{Let } 7^x = y$$

$$-y^2 + 10y + 2 = 0$$

$$y = \frac{-10 \pm \sqrt{10^2 - 4(-1)(2)}}{2(-1)}$$

$$y = \frac{-10 \pm \sqrt{108}}{-2}$$

$$y = \frac{-10 \pm 6\sqrt{3}}{-2}$$

$$y = 5 \pm 3\sqrt{3}$$

$$7^x = 5 \pm 3\sqrt{3}$$

$$7^x = 5 + 3\sqrt{3} \text{ or } 7^x = 5 - 3\sqrt{3}$$

$$x = \log_7(5 + 3\sqrt{3}) \text{ or } x = \log_7(5 - 3\sqrt{3})$$

$$x \approx 1.1933$$